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# SQL

Structured query language if is a standard programming language specifically designed for managing and manipulate, relational database. It allocos users to create, read, ~~edit~~ update and delete data stored in a database

## Data

Data refers raw unprocessed facts and figures that have no context or meaning on their ocon. In databases data can be numbers texts, dates, images or any other type of informations stored systematically. When data is processed and given contexts it become meaningful information.

Ex: Sanjana, 23/10/2005, 20

## Database.

A database is a organised collection of structured data stored electronically and designed for effecient ~~redrival~~ retrieval data insertion and management.

It act as a central repository that allocos



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multiple users and application to access shared data in a controlled and a consistant manner

Syntax

(i) Create : used to create a new database

Sy: CREATE DATABASE database_name;

Ex: CREATE DATABASE college;

(ii) Access: used to select/ access the database

Sy: USe database_name;

Ex: USE college;

(iii) Views : used to display all available database

Sy: SHOW DATABASES;

Ex: SHOW DATABASES

(iv) Delete : used to permanently delete a database.

Sy: DROP DATABASE database_name;

Ex. DROP DATABASE college;



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# DBMS

A Database management system is a software that provides an interface b/w user or application and database.
It handles data storage, retrival, security and integrity. it elimánate the complexity of managing raw datafiles offering tools to define manipulate and control data effecientley

Ex: MySQL - opensourse RDBMS
Oracle - enterprise grade DBMS
SQL server - microsoffs DBMS

## Types of DBMS

(i) HDBMS (Hierarchical)
Datas are stored in tree format
Ex: IBM IMS

(ii) NDBMS (Network)
Data are stored in graphs
Ex: IDMS

(iii) ODBMS (object oriented)
stores datas in the forms of objects.
Ex: db4o



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(iv) RDBMS.

datas are stored in the forms of rows and columns. It is most world wide language used
Ex: MySQL, Oracle

## Objectives of SQL

1. Mass Storage
2. Data redudancy
3. Data protection
4. platform independence
5. Improved data sharing.

## History of SQL

* In 1970 Dr EF Codd proposed the relational model at IBM.
* In 1974 SQL was developed by Donald D Chamberlia and Raymond F Boyce at IBM later in 1979 oracle became a first commercial SQL product.
* 1986 SQL became ANSI standardise approve



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1. What is ER Diagram

ER = Entity - Relationship . An ER diagram is a visual blueprint of a database . We use it before creating the actual database to understand what data we need and how different data are connected.

Think of it like : Real world system → ER diagram → database.

Ex : for a college database, we may have student, faculty, course, department, Result.

Components of ER model -

(i) Entity

An entity is a real world object that we want to store information about.

It is represented by a rectangle [ ]

Ex : Student is an entity , faculty, course, department, result.

★ Entity = who or what are we storing data about.



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(ii) Attribute

An attribute describes the properties or characteristics of an entity.
It is represented by an oval ( )

Ex: The entity student can have
student = Roll no, name, branch, semester, mobile no., address, age
so, entity = students.
Attribute = student details.

Types of attribute

a. Single valued attribute.

An attribute that has only one value for an entity
Ex: student Roll no → 101
Student has only 1 roll number
Represented as oval ( )

b. Multi valued Attribute.

An attribute that can have multiple values for same entity
Ex: student → phone no.
One student can have multiple phone no.



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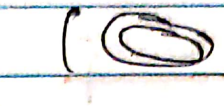
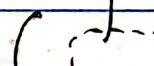
represented by double oval (  )

c. Derived attribute

an attribute whose value is calculated from another attribute

Ex: Date of birth → Age.

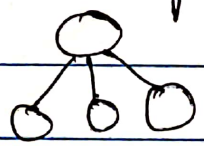
* if DOB is stored, the system can calculate age
* Age is calculated, not directly stored.
* represented by Dashed oval (  )

d. Composite Attribute

An attribute that can be divided into smaller sub parts

Ex: Name → first name
→ Middle name
→ Last name

Name is made up of multiple smaller attribute

Represented as 

e. Stored attribute.

An attribute whose value is directly stored in the database

Ex: DOB = 10/03/2005

DOB is entered and stored directly in the DB

represented as oval ()



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(iii) Relationship

A relationship shows the association b/w 2 or more entities.

* It is represented by a diamond (◇) and placed b/w entities, with lines connecting them
* Student – Issues – Book.

Here: student = Entity
Book = Entity
Issues = Relationship

A student issues a book.

Ex: In a college
student = entity
Name, Rollno, branch = attribute
course = entity
Enrolls = Relationship

## Data Types

SQL data types define the kind of values a column can hold, choosing the right data type ensures data integrity, storage efficiency and query performance.



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# Types of Data types.

## 1. Numeric Data types.

→ INT / INTEGER : whole numbers

Eg : 1, 100, -50

→ FLOAT / DOUBLE : Decimal numbers

Eg : 3.14, 99.99

→ DECIMAL (p,s) : used when you need accurate decimal values, especially for money.

Ex : DECIMAL (10,2) for currency

→ BIGINT : used for very large whole no.

Ex : population BIGINT

population of a large country, very large transaction ID's, etc

## 2. String Data type.

→ CHAR(n) : fixed - length string, always uses n bytes.

Ex: CHAR (10)

→ VARCHAR (n) : variable length string, uses actual length

Ex : VARCHAR (255)

imp point is that it uses storage based on the actual length of the value, rather than always using the full declared length.

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→ TEXT : Large text content (upto 65,535 CHARS)
Eg : Address

→ ENUM : allows to select one value from a predefined list
Ex : gender ENUM ('Male', 'Female', 'others')

3. Date / Time Data types.

→ DATE : stores date only (F): (YYYY-MM-DD)
Ex : Joining_date DATE

→ TIME : stores time only (F): (HH:MM:SS)
Ex : 09:30:00

→ DATETIME : stores both date and time
Ex : 2026-09-22 09:05:25

→ YEAR : stores 4-digit year
Ex : 2026

→ MONTH : stores only month values upto 2 digit
Ex : 01, 09, 12

## Sub language of SQL

SQL is divided into 5 sub language, each serving a distinct purpose in database operations. understanding these categories is fundamental to mastering SQL



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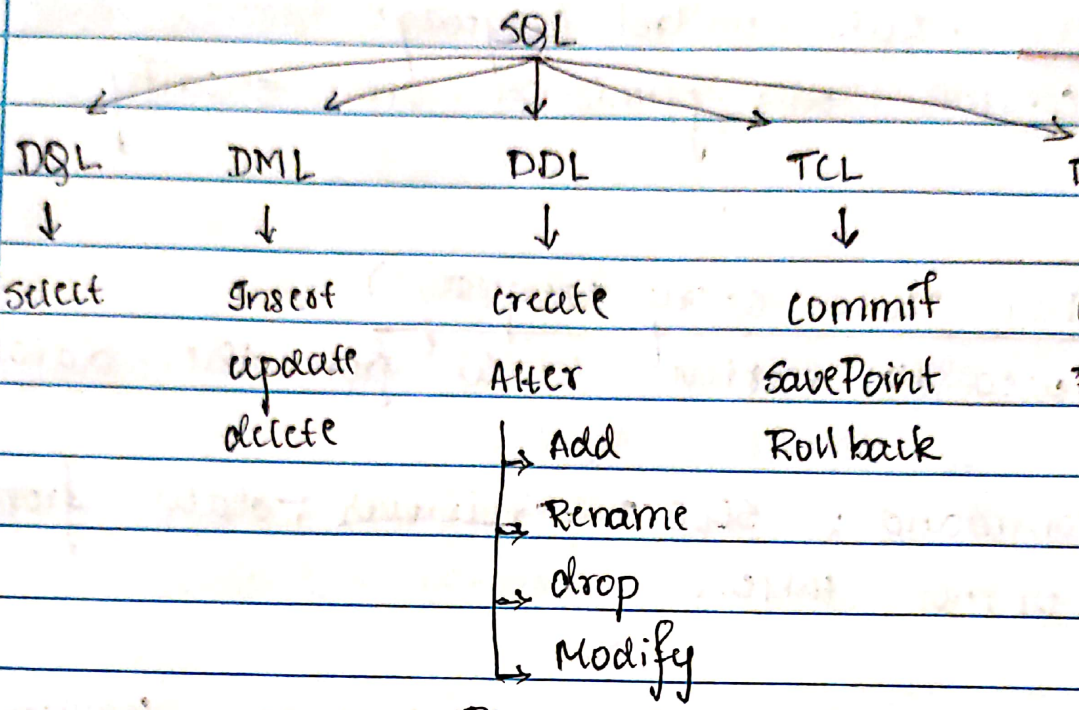


1. DQL - Data ~~definition~~ Query language
Retrieves data from the database

2. DML - Data manipulation Language
Manipulates data stored in tables.

3. DDL - Data definition Language
Defines and modifies the database structure

4. TCL - Transaction control language
manages transactions to ensure data consistency



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5. DCL - Data control language.
controls user permission and security

1. DQL (Data Query language)
used to retrieve data from the database

command ; SELECT = retrieves data from one or more tables.

Syntax : SELECT * FROM table_name ; (retrieves entire data)
SELECT column_name FROM table_name ;
(retrieves only particular column data)

Ex : SELECT * FROM employee ;
SELECT name FROM employee ; (retrives only name column)
SELECT name, salary FROM employee ; (retrives multiple columns from table)

2. DML (Data manipulation Language)
This is used to manipulate (change the data stored inside tables
also called as a 3GL level language

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* These changes can be rolledback using TCL command like ROLLBACK before being committed

Commands.

a. Insert : adds new row/ record to a table.

Syntax : INSERT INTO table_name (c1, c2, c3)
VALUES (v1, v2, v3)

Ex: INSERT INTO employee (id, name, salary, dept)
VALUES (4, 'sanjana', 25,000, 'Finance');

b. UPDATE : modifies existing data in a table.

Syntax : UPDATE table_name
SET column1 = value1,
column2 = value2
WHERE condition;

Suppose Sanjana's salary changes from 30,000 to 35,000

Ex: UPDATE employee
SET salary = 35,000
WHERE name = 'sanjana';

IMP
UPDATE employee
SET salary = 35,000;
can update every employee salary. So use WHERE



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c DELETE : delete removes rows from a table based on a condition.

syntax: DELETE FROM table_name
WHERE condition ;

Ex: DELETE FROM employee.
WHERE name = 'sanjana' ;
sanjana's entire row will be removed.

* Be careful with DELETE FROM employee;
This removes all ~~data~~ row from the table.

3. DDL (Data definition language)
used to define and modify the structure of the ~~data~~database objects.
DML changes the data
DDL changes the structure
DDL is column level language

a. CREATE : is used to create a database object such as a tabel
syntax: CREATE TABEL tabel_name(
column1 datatype CONSTRAINTS,
column2 datatype,
column3 datatype
);

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Ex: CREATE TABEL employeed
id INT,
name VARCHAR (50),
salary INT
);

This creates a tabel called employee

employee
id | name | salary

There are no records yet.

B. ALTER : This modifies an existing database object / tabel structure

There are several operations under ALERT.

(i) ALERT +ADD : adds a new column

syntax: ALERT TABLE table_name
ADD column_name datatype;

Ex: ALERT TABEL employee
ADD email VARCHAR (100);

id | name | salary | email

This adds a new column.

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(ii) ALTER + RENAME : changes a column name.

Syntax : ALTER TABEL table_name
RENAME COLUMN old_column_name
TO new_column_name.

Ex: ALTER TABLE employee.
RENAME COLUMN name
TO employee_name;

(iii) ALTER + MODIFY : It modifies either the data type of the existing column when there is no data present in the specific column
* or helps in changing the position of the column in the table

syntax : ALTER TABLE table_name
MODIFY column_name new_datatype;

Ex: ALTER TABEL employee
MODIFY salary DECIMAL (10,2);
changes the datatype of salary.

Syntax : ALTER TABLE table_name
MODIFY column_name old_datatype
FIRST;



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or ALTER TABEL tabel-name
MODIFY column_name old datatype;
AFTER column_name.

Ex: ALTER TABLE employee
MODIFY salary INT FIRST;

~~ALTER TABLE employee~~
Before: id | name | salary | department
After: salary | id | name | department

Ex: ALERT TABLE employee
MODIFY salary INT AFTER name;

before: salary | id | name | department
after: id | name | salary | department

Ex: ALTER TABEL employee
MODIFY department VARCHAR(50) AFTER name;

id | name | department | salary.



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(iv) ALTER + DROP : Removes a column completely

syntax: ALTER tabel tabel_name
DROP COLUMN column-name;

Ex: ALTER TABLE employee
DROP COLUMN email;
The email column is no longer exists.

c. ~~(iv)~~ Rename : changes the name of a database object, such as table.

syntax: RENAME TABEL old_tabel_name
TO new-table-name;

Ex: RENAME TABEL employee
TO employees;

d. Drop : removes the table / object itself, including its structure.
deleting the tabel / database object completely

syntax : DROP TABEL tabel-name;

Ex: DROP TABEL employee;
Now the tabel itself gone.



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Before : employee

| id | name | salary |
|---|---|---|

after : employee x

e TRUNCATE : removes all rows from the tabel but keeps the tabel structure.

syntax : TRUNCATE TABEL tabel_name;

Ex : TRUNCATE TABEL employee;

Before :

| id | name | salary |
|---|---|---|
| 1 | sana | 25,000 |
| 2 | sanju | 30,000 |

after :

| id | name | salary |
|---|---|---|

DELETE : → selected rows or all rows is removed
→ tabel structure remains
→ WHERE condition is used

TRUNCATE : → All rows is removed
→ tabel structure remains
→ NO WHERE condition

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DROP : → Entire table is removed
→ table structure is removed
→ NO WHERE condition.

4. TCL (Transaction control language)

TCL is used to control transactions.
manage transactions in a database and helps maintain data consistency

A transaction is a group of database operations treated as a unit.

Ex: UPDATE employee
SET salary = 35000
WHERE id = 1';

This is a change made during a transaction
TCL allows us to decide whether to save the change or undo it.

Types

1. COMMIT
2. ROLLBACK
3. SAVEPOINT



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1) WHERE Clause

The WHERE clause is used to filter records from a table.

It tells SQL Give me only the rows that sp satisfy this condition.

Without WHERE, all rows are returned or affected

Syntax: SELECT * FROM table_name
WHERE condition;

Ex: SELECT * FROM employees
WHERE dept = 'sales';

This filters only sales rows.

2) OPERATORS

Operators are symbols or keywords used to perform operations on values in SQL

There are 6 Types:

1. Arithmetic operators.

→ addition (+)
→ subtraction (-)
→ multiplication (*)



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→ Division (/)

→ Modulus (%) the returns reminder

Ex: SELECT salary + 5000
FROM employees;

if salary = 25000

Result = 30000

2. Relative / comparison operators

used to compare 2 values

→ (=) equal to

→ (!= or <>) Not equal to

→ (<) Less than

→ (>) Greater than

→ (<=) Lessthan or equal to

→ (>=) Greater than or equal to.

Ex: SELECT * FROM employees.
WHERE salary > 25000;

This returns employees whose salary is greater than £25,000

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3. Logical operator
used to combine multiple conditions

→ AND : Bothe condition must be TRUE

```
SELECT * FROM employees
WHERE dept = 'sales' AND salary > 20000;
```

department must be sales AND salary must be greater than 20,000

→ OR : At least one condition must be TRUE

```
SELECT * FROM employees
WHERE dept = 'sales' OR dept = 'IT';
```

Department can be Sales or IT

→ NOT : means opposite / excleude the condition

```
SELECT * FROM employees
WHERE NOT dept = 'sales';
```

Give employees who are not in Sales.



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4. Set operators

used to combine results from queries.

→ UNION : combines results and removes duplicates

```
SELECT name FROM employees
WHERE dept = 'sales'
UNION
SELECT name FROM employees
WHERE salary > 25000;
```

→ UNION ALL : combines results and keeps duplicates

```
SELECT name FROM employees
WHERE dept = 'Sales'
UNION ALL
SELECT name FROM employees
WHERE salary > 25000;
```

5. Special operators.

→ IN : used when we want to check multiple possible values.

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```
SELECT * FROM employees
WHERE dept IN ('sales', 'IT');
```

→ BETWEEN : used to check whether a value falls within a specific range

```
SELECT * FROM employees
WHERE salary BETWEEN 20000 AND 30000;
```

→ LIKE : used for pattern matching.

```
SELECT * FROM employees
WHERE name LIKE 's%';
```

% means any no. of characters. (s%)
so this can find names starting with s.

(%s) → searchs names ends with s
(%s%) → searchs names ~~which~~ which contains s anywhere.

→ IS NULL : used to check for null values.

```
SELECT * FROM employees
WHERE salary IS NULL;
```

This returns emp whose salary is missing.



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→ AS : Used to give a temporary / alias name to a column or table.

```
SELECT salary AS monthly_salary
FROM employees;
```

Here monthly_salary is temporary name displayed for salary.

3 Literals.

Literals is an fixed value written directly in a SQL query. It doesn't change during query execution

There are 3 types of literals.

→ String Literals.

Text values are generally written inside quotes.

```
SELECT * FROM employees
WHERE dept = 'sales';
```

'sales' → string literals.



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→ **Numerical literal**

No. written directly in a query.

```
SELECT * FROM employees
WHERE salary > 50000 ;
```

50000 → Number literals

→ **Date Literals**

A date value written directly

```
SELECT * FROM employees
WHERE doj = '2026-08-21';
```

'2026-08-21' → Date Literal

**Literal vs Variable**

| Literal | Variable |
|---|---|
| * fixed values | * can change |
| * written directly | * Stored temporarily |

4. **Dual Table.**

Dual is a special dummy table used when we want to execute a SELECT without retrieving data from an actual table.
It is a dummy table created for practice
It contains only 1 row & 1 column.



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Syntax : SELECT data FROM DUAL;
SELECT data;

Ex: SELECT 10+20 FROM DUAL;
O/P : 30
Ex: SELECT 'HELLO' FROM DUAL;
O/P : HELLO.

NOTE: operators control logic,
Literals provides values,
Dual helps us practice.

5 TRIM

This Removes Leading and or trailing spaces from a string (or specified char) from a string.

(LTRIM) → Removes from the left
(RTRIM) → removes from right
(TRIM) → removes from both sides

TRIM Ex: SELECT TRIM(' HELLO ') AS trimmed;
O/P: HELLO

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LTRIM Ex: SELECT LTRIM ('  SQL') AS left_trimmed;
o/p : SQL

RTRIM Ex: SELECT RTRIM('SQL  ') As right_trimmed;
o/p : SQL

6. Distinct

This removes duplicate values / rows from the results and returns unique values

Syntax: SELECT DISTINCT column_name
FROM table_name;

Ex: dept
sales
IT
sales
HR
IT

SELECT DISTINT dept FROM employees; → dept
sales
IT
HR



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# CONSTRAINTS.

Constraints are rules applied to columns in a table. They help make sure that the data entered into the database is

→ correct, valid, reliable, consistent

They prevent invalid data from being inserted into a tabel

Ex: suppose we have employee table.

| ID | Name | Age |
|---|---|---|
| 001 | sanju | 22 |
| 002 | sand | 25 |

if we say Age must be greater than 18, SQL can prevent someone from entering Age <10. That rule is constraint.

## Types of constraints.

a. **Unique.**

ensures that all values in a column are different.



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Ex: emp email ID must be different

```
CREATE TABEL Employee (
    email VARCHAR (100) UNIQUE
);
```

1000, sanjana@gmail.com
sanjana@gmail.com
The second value will not be allowed bez if is depricate.

→ This ensures all values in a column are different unlike Primary key
→ a table can have multiple unique constraints like.

```
CREATE TABEL Employee (
    emcid VARCHAR (100) UNIQUE
    name VARCHAR (20) UNIQUE
);
```

→ And UNIQUE columns can contain NULL (one NULL per column.

B. Not NULL

The column should not be empty (NULL. Every row must provide a value for this column



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Ex: CREATE TABEL Employees (
    name VARCHAR (50) NOT NULL
);

c. CHECK

allows us to specify a condition that the date must satisfy

Ex: CREATE TABEL Employee (
    age INT CHECK( age >= 18)
);

d. DEFAULT.

DEFAULT automatically provides a value when we don't give one during insertion, INSERT

CREATE TABLE Employee (
    name VARCHAR (50),
    city VARCHAR (30) DEFAULT 'Bangalore'
);

e. PRIMARY KEY

uniquely identifies each row in a table. combination of not null and unique. Each table can have only one primary key



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```
CREATE TABEL EMPLOYEE (
    employee_id INT PRIMARY KEY,
    name VARCHAR(50)
);
```

F. FOREGIN KEY.

Creates a link blw 2 tables by referencing the primary key of another table.

It enforces referential integrity. ensures that value in a column match values in the referenced table

Ex: Imagin we have 2 tables

Table 1: Employee

| employee_id | name |
|---|---|
| 101 | sanjana |
| 102 | Rama |
| 103 | Sana |

Here, employee_id is the primary key

Table 2: Attendance.

| attendance_id | employee_id | status |
|---|---|---|
| 1 | 101 | present |
| 2 | 102 | Absent. |

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Here, employee_id is the ~~prima~~ foredin key

It refers to : Attendance.employee_id
↓
Employee.employee_id

```
CREATE TABEL Attendance (
    Attendance_id INT PRIMARY KEY,
    employee_id INT,
    statues VARCHAR(20),
    FOREGIN KEY (employee_id)
    REFERENCES Employee (employee_id)
);
```



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